

Task 1.

Part 1: Normalization

Task 1: Identification of Normal Form Violations

Original Table:

Booking_receipt (booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)

Answer:

The original table violates the **First Normal Form - 1NF**.

Reason:

- 1. **Non-atomic values:** The seat_numbers column contains comma separated values (like, "12A, 12B"), which are non-atomic, thus, divisible.
- 2. **Repeating groups:** When a single booking covers multiple passengers, the passenger attributes (passenger_full_name, passenger_passport_number) act as repeating groups per booking_id. That means : if a table has a line with an id of 1n, and this line contains attributes described earlier, it will NOT violate 1NF if there is only one passenger in this line, BUT if multiple passengers were written with this single id of 1, the table cannot easily query, update, or index multiple values stuffed into a single column.

Tasks 2 & 3: Step-by-Step Normalization & Functional Dependencies (FDs)

Step 1: 1NF (First Normal Form)

To convert the table to 1NF, repeating lists are split into individual rows. Each row represents a single passenger and seat combination. The primary key becomes a composite key: (booking_id, passenger_passport_number).

Functional Dependencies (FDs):

- 1. booking_id, passenger_passport_number → passenger_full_name, seat_number
- 2. booking_id → flight_number, ticket_price
- 3. flight_number → departure_airport_name, arrival_airport_name, airline_name
- 4. departure_airport_name → departure_city
- 5. arrival_airport_name → arrival_city

Resulting 1NF Table:

- Booking_receipt_1NF (booking_id, passenger_passport_number, passenger_full_name, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_number, ticket_price)

Step 2: 2NF (Second Normal Form)

To convert to 2NF, we eliminate **partial dependencies** (attributes depending on only part of the composite key):

- passenger_full_name depends solely on passenger_passport_number.
- Flight information and pricing depend solely on booking_id.

Resulting 2NF Tables:

1. **Passengers:** passenger_passport_number, passenger_full_name
2. **Bookings:** booking_id, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price
3. **Booking_Passengers:** booking_id, passenger_passport_number, seat_number

(Foreign Keys: booking_id → Bookings, passenger_passport_number → Passengers)

Step 3: 3NF (Third Normal Form)

To convert to 3NF, we eliminate **transitive dependencies** (non-key attributes depending on other non-key attributes):

- In Bookings: flight_number determines departure_airport_name, arrival_airport_name, and airline_name.
- departure_airport_name determines departure_city.
- arrival_airport_name determines arrival_city.

Final 3NF Schema:

1. **Passengers:** passenger_id, passenger_full_name, passenger_passport_number)
2. **Airports:** airport_name, city
3. **Airlines:** airline_name
4. **Flights:** (flight_number, departure_airport_name, arrival_airport_name, airline_name

(FKs: departure_airport_name → Airports, arrival_airport_name → Airports, airline_name → Airlines)

5. **Bookings:** (booking_id, flight_number, ticket_price)

(FK: flight_number → Flights)

6. **Booking_Passengers:** booking_id, passenger_id, seat_number

(FKs: booking_id → Bookings, passenger_id → Passengers)

Task 4: 3NF Verification for DDL Flights Table

- **Primary Key:** flight_id
- **Functional Dependencies:** flight_id → all non-key attributes.
- **Analysis:** The Flights table contains foreign keys (departing_airport_id, arriving_airport_id, airline_id) pointing to separate entity tables. Descriptive metadata (like airport names or cities) lives in Airport and Airline tables. Thus, all non-key attributes depend directly on flight_id.

- **Conclusion: Yes, the Flights table is in 3NF.** No transitive dependencies exist.

Task 5: Implementing Task 2 Tables and Data Insertion

```
CREATE TABLE Airports ( airport_name VARCHAR(50) PRIMARY KEY, city
VARCHAR(50) NOT NULL );
```

```
CREATE TABLE Airlines ( airline_name VARCHAR(50) PRIMARY KEY );
```

```
CREATE TABLE Flights ( flight_number VARCHAR(20) PRIMARY KEY,
departure_airport_name VARCHAR(50) NOT NULL, arrival_airport_name
VARCHAR(50) NOT NULL, airline_name VARCHAR(50) NOT NULL, FOREIGN
KEY (departure_airport_name) REFERENCES Airports(airport_name),
FOREIGN KEY (arrival_airport_name) REFERENCES Airports(airport_name),
FOREIGN KEY (airline_name) REFERENCES Airlines(airline_name) );
```

```
CREATE TABLE Passengers ( passenger_id INT PRIMARY KEY,
passenger_full_name VARCHAR(100) NOT NULL,
passenger_passport_number VARCHAR(20) NOT NULL UNIQUE );
```

```
CREATE TABLE Bookings ( booking_id INT PRIMARY KEY, flight_number
VARCHAR(20) NOT NULL, ticket_price DECIMAL(7,2) NOT NULL, FOREIGN KEY
(flight_number) REFERENCES Flights(flight_number) );
```

```
CREATE TABLE Booking_Passengers ( booking_id INT NOT NULL, passenger_id
INT NOT NULL, seat_number VARCHAR(10) NOT NULL, PRIMARY KEY
(booking_id, passenger_id), FOREIGN KEY (booking_id) REFERENCES
Bookings(booking_id), FOREIGN KEY (passenger_id) REFERENCES
Passengers(passenger_id) );
```

```
INSERT INTO Airports VALUES ('JFK International', 'New York'), ('Heathrow',
'London'), ('Haneda', 'Tokyo'), ('Charles de Gaulle', 'Paris'), ('Dubai
International', 'Dubai');
```

```
INSERT INTO Airlines VALUES ('Delta Air Lines'), ('British Airways'), ('Japan
Airlines'), ('Air France'), ('Emirates');
```

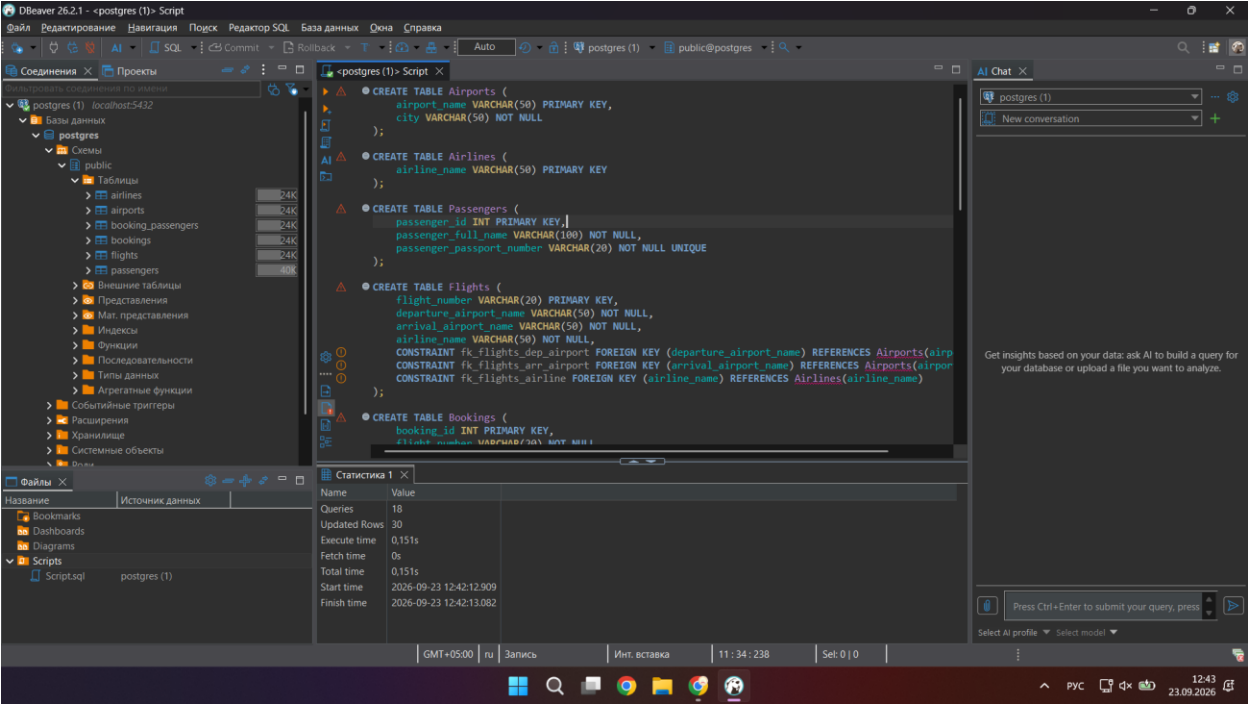
```
INSERT INTO Flights VALUES ('DL100', 'JFK International', 'Heathrow', 'Delta Air
Lines'), ('BA200', 'Heathrow', 'Haneda', 'British Airways'), ('JL300', 'Haneda',
'Charles de Gaulle', 'Japan Airlines'), ('AF400', 'Charles de Gaulle', 'Dubai
International', 'Air France'), ('EK500', 'Dubai International', 'JFK International',
'Emirates');
```

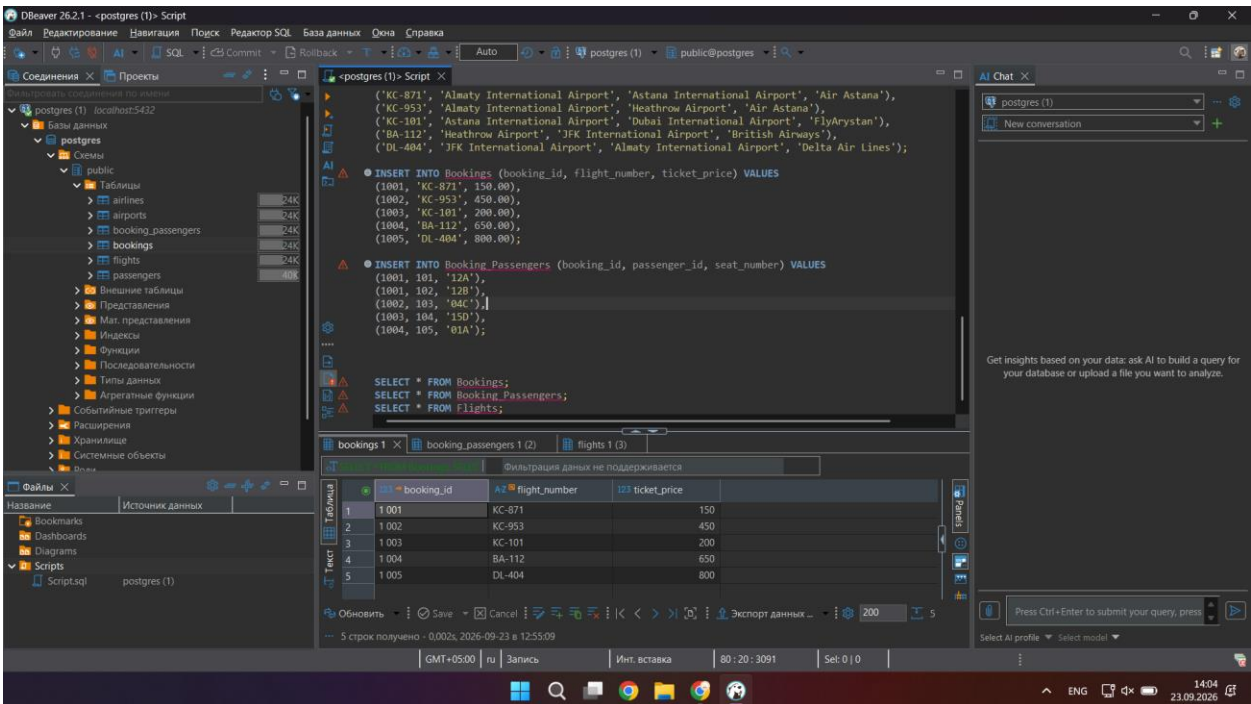
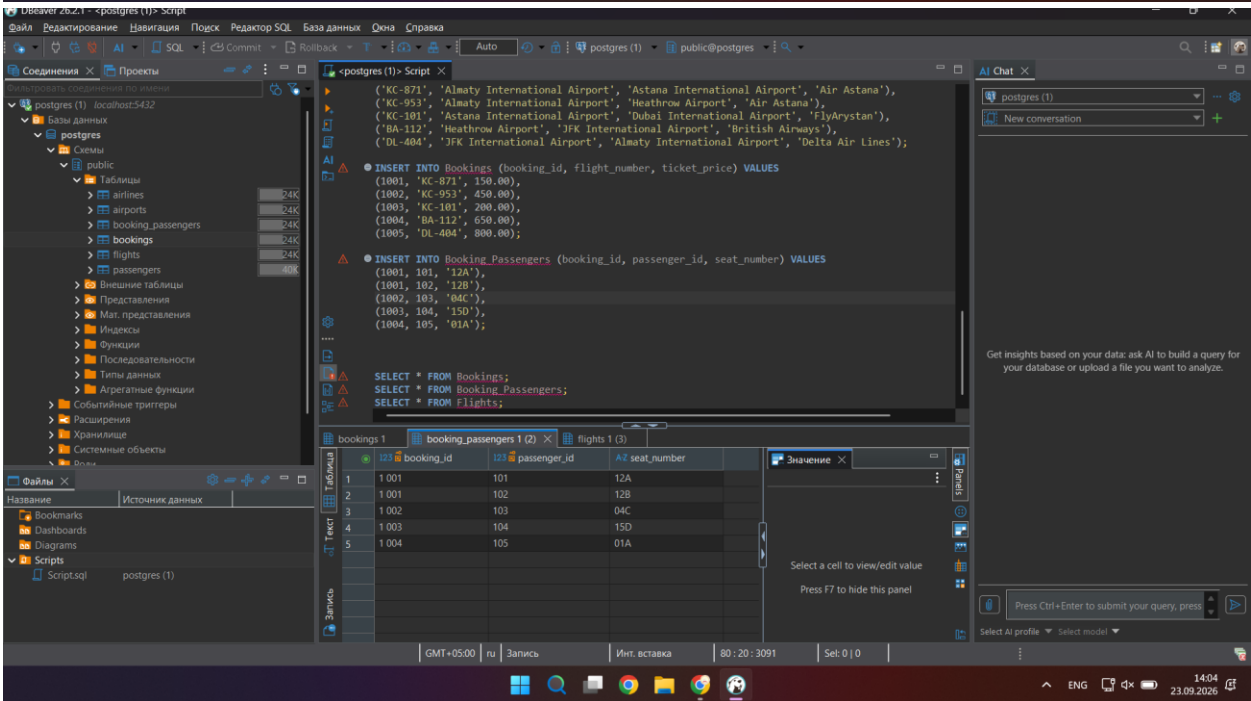
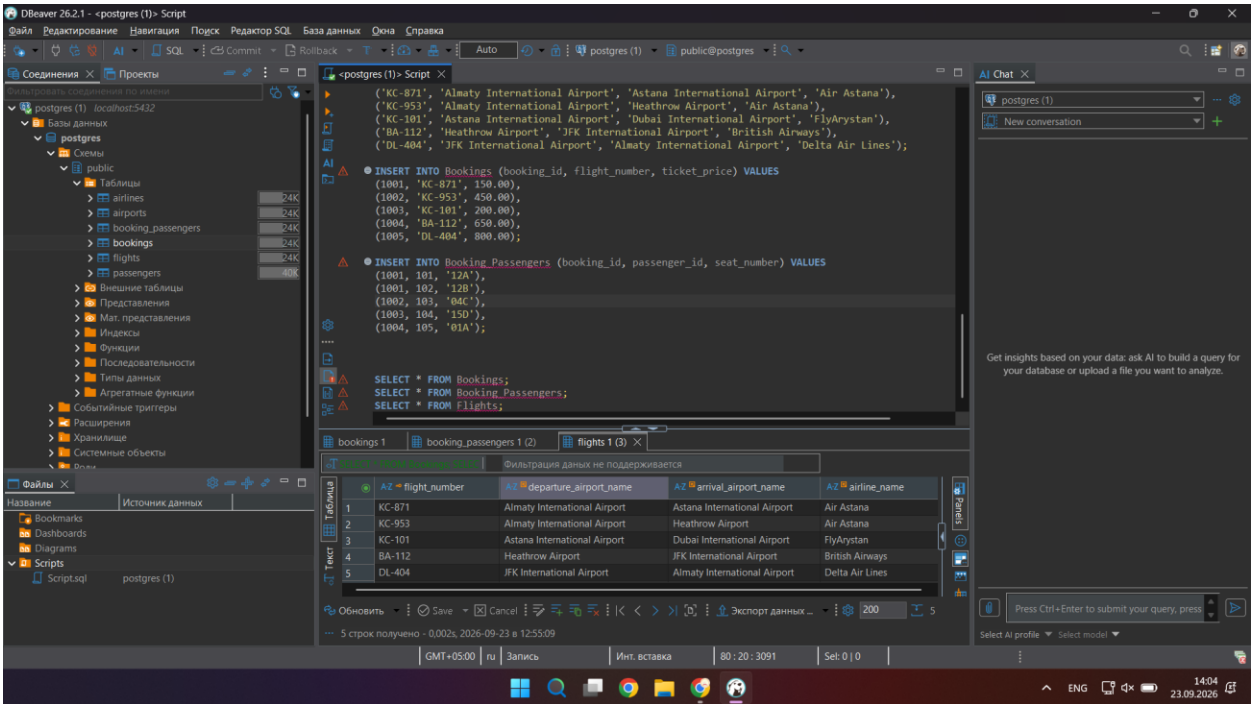
```
INSERT INTO Passengers VALUES (1, 'John Doe', 'A12345678'), (2, 'Jane Smith',
'B87654321'), (3, 'Alice Johnson', 'C11223344'), (4, 'Bob Brown', 'D44332211'),
(5, 'Charlie Davis', 'E55667788');
```

```
INSERT INTO Bookings VALUES (101, 'DL100', 450.00), (102, 'BA200', 600.00),
(103, 'JL300', 750.00), (104, 'AF400', 520.00), (105, 'EK500', 800.00);
```

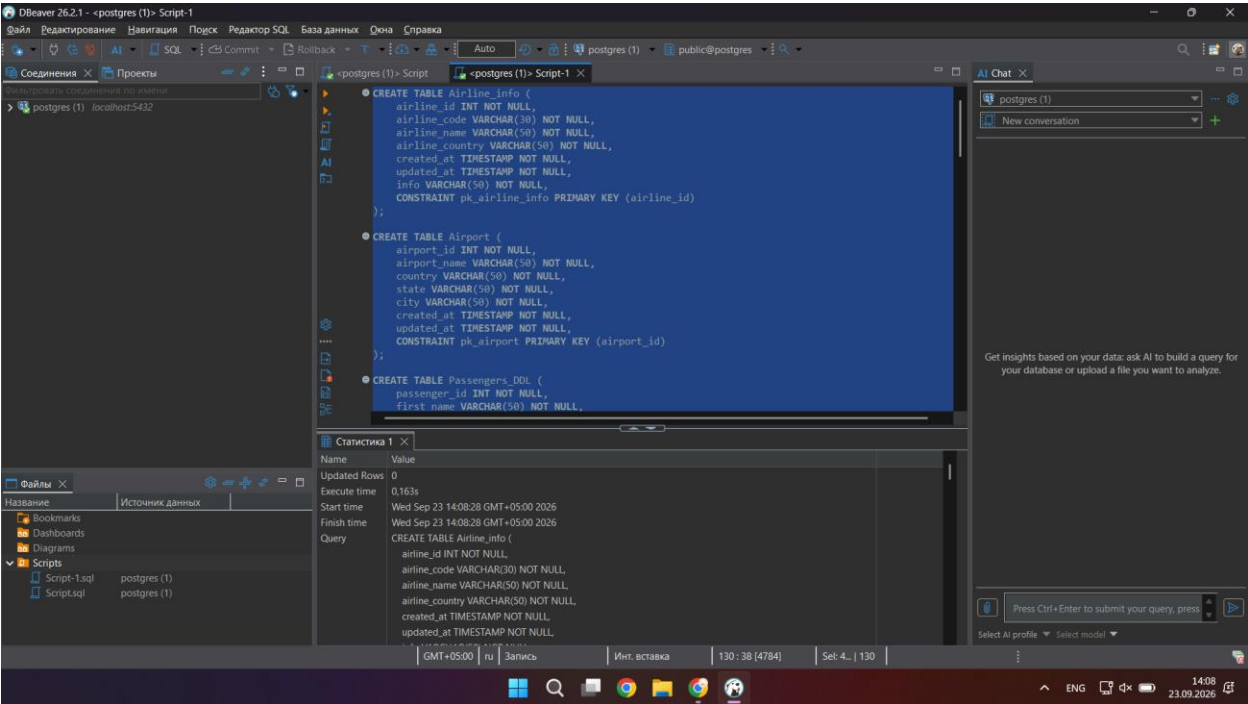
```
INSERT INTO Booking_Passengers VALUES (101, 1, '12A'), (102, 2, '14B'), (103,
3, '01A'), (104, 4, '22C'), (105, 5, '05D');
```

In the following screenshots, we can clearly see that script has successfully executed and has 0 errors. Besides that, the DB is not updated, so the errors with this description: table ** not found is ONLY because the cache of the DB cannot see that all the necessary table DO exist.





DDL



Scripts :

```
CREATE TABLE Airports (  
airport_name VARCHAR(50) PRIMARY KEY,  
city VARCHAR(50) NOT NULL  
);  
  
CREATE TABLE Airlines (  
airline_name VARCHAR(50) PRIMARY KEY  
);  
  
CREATE TABLE Passengers (  
passenger_id INT PRIMARY KEY,  
passenger_full_name VARCHAR(100) NOT NULL,  
passenger_passport_number VARCHAR(20) NOT NULL UNIQUE  
);  
  
CREATE TABLE Flights (  
flight_number VARCHAR(20) PRIMARY KEY,  
departure_airport_name VARCHAR(50) NOT NULL,  
arrival_airport_name VARCHAR(50) NOT NULL,  
airline_name VARCHAR(50) NOT NULL,  
CONSTRAINT fk_flights_dep_airport FOREIGN KEY (departure_airport_name) REFERENCES  
Airports(airport_name),  
CONSTRAINT fk_flights_arr_airport FOREIGN KEY (arrival_airport_name) REFERENCES  
Airports(airport_name),  
CONSTRAINT fk_flights_airline FOREIGN KEY (airline_name) REFERENCES  
Airlines(airline_name)  
);  
  
CREATE TABLE Bookings (  
booking_id INT PRIMARY KEY,  
flight_number VARCHAR(20) NOT NULL,  
ticket_price DECIMAL(7, 2) NOT NULL,  
CONSTRAINT fk_bookings_flight FOREIGN KEY (flight_number) REFERENCES  
Flights(flight_number)  
);  
  
CREATE TABLE Booking_Passengers (  
booking_id INT NOT NULL,
```

```
passenger_id INT NOT NULL,  
seat_number VARCHAR(10) NOT NULL,  
PRIMARY KEY (booking_id, passenger_id),  
CONSTRAINT fk_bp_booking FOREIGN KEY (booking_id) REFERENCES Bookings(booking_id),  
CONSTRAINT fk_bp_passenger FOREIGN KEY (passenger_id) REFERENCES  
Passengers(passenger_id)  
);  
  
INSERT INTO Airports (airport_name, city) VALUES  
( 'Astana International Airport', 'Astana'),  
( 'Almaty International Airport', 'Almaty'),  
( 'Heathrow Airport', 'London'),  
( 'JFK International Airport', 'New York'),  
( 'Dubai International Airport', 'Dubai');  
  
INSERT INTO Airlines (airline_name) VALUES  
( 'Air Astana'),  
( 'FlyArystan'),  
( 'British Airways'),  
( 'Delta Air Lines'),  
( 'Emirates');  
  
INSERT INTO Passengers (passenger_id, passenger_full_name, passenger_passport_number)  
VALUES  
(101, 'Abay Orazbekuly', 'N12345678'),  
(102, 'Alihan Bokeikhanov', 'N87654321'),  
(103, 'Akhmet Baitursynov', 'N11223344'),  
(104, 'Mirzhakip Dulatov', 'N55667788'),  
(105, 'Magzhan Zhumabayev', 'N99887766');  
  
INSERT INTO Flights (flight_number, departure_airport_name, arrival_airport_name,  
airline_name) VALUES  
( 'KC-871', 'Almaty International Airport', 'Astana International Airport', 'Air  
Astana'),  
( 'KC-953', 'Almaty International Airport', 'Heathrow Airport', 'Air Astana'),  
( 'KC-101', 'Astana International Airport', 'Dubai International Airport', 'FlyArystan'),  
( 'BA-112', 'Heathrow Airport', 'JFK International Airport', 'British Airways'),  
( 'DL-404', 'JFK International Airport', 'Almaty International Airport', 'Delta Air  
Lines');  
  
INSERT INTO Bookings (booking_id, flight_number, ticket_price) VALUES  
(1001, 'KC-871', 150.00),  
(1002, 'KC-953', 450.00),  
(1003, 'KC-101', 200.00),  
(1004, 'BA-112', 650.00),  
(1005, 'DL-404', 800.00);  
  
INSERT INTO Booking_Passengers (booking_id, passenger_id, seat_number) VALUES  
(1001, 101, '12A'),  
(1001, 102, '12B'),  
(1002, 103, '04C'),  
(1003, 104, '15D'),  
(1004, 105, '01A');  
  
SELECT * FROM Bookings;  
SELECT * FROM Booking_Passengers;  
SELECT * FROM Flights;
```

DDL:

```
CREATE TABLE Airline_info (  
  airline_id INT NOT NULL,  
  airline_code VARCHAR(30) NOT NULL,  
  airline_name VARCHAR(50) NOT NULL,  
  airline_country VARCHAR(50) NOT NULL,  
  created_at TIMESTAMP NOT NULL,  
  updated_at TIMESTAMP NOT NULL,  
  info VARCHAR(50) NOT NULL,  
  CONSTRAINT pk_airline_info PRIMARY KEY (airline_id)  
);  
  
CREATE TABLE Airport (  
  airport_id INT NOT NULL,  
  airport_name VARCHAR(50) NOT NULL,  
  country VARCHAR(50) NOT NULL,  
  state VARCHAR(50) NOT NULL,  
  city VARCHAR(50) NOT NULL,  
  created_at TIMESTAMP NOT NULL,  
  updated_at TIMESTAMP NOT NULL,  
  CONSTRAINT pk_airport PRIMARY KEY (airport_id)  
);  
  
CREATE TABLE Passengers_DDL (  
  passenger_id INT NOT NULL,  
  first_name VARCHAR(50) NOT NULL,  
  last_name VARCHAR(50) NOT NULL,  
  date_of_birth DATE NOT NULL,  
  gender VARCHAR(50) NOT NULL,  
  country_of_citizenship VARCHAR(50) NOT NULL,  
  country_of_residence VARCHAR(50) NOT NULL,  
  passport_number VARCHAR(20) NOT NULL,  
  created_at TIMESTAMP NOT NULL,  
  updated_at TIMESTAMP NOT NULL,  
  CONSTRAINT pk_passengers_ddl PRIMARY KEY (passenger_id)  
);  
  
CREATE TABLE Security_check (  
  security_check_id INT NOT NULL,  
  check_result VARCHAR(20) NOT NULL,  
  created_at TIMESTAMP NOT NULL,  
  updated_at TIMESTAMP NOT NULL,  
  passenger_id INT NOT NULL,  
  CONSTRAINT pk_security_check PRIMARY KEY (security_check_id),  
  CONSTRAINT fk_sec_passenger FOREIGN KEY (passenger_id) REFERENCES  
  Passengers_DDL(passenger_id)  
);  
  
CREATE TABLE Booking_DDL (  
  booking_id INT NOT NULL,  
  flight_id INT NOT NULL,  
  passenger_id INT NOT NULL,  
  booking_platform VARCHAR(50) NOT NULL,  
  created_at TIMESTAMP NOT NULL,  
  updated_at TIMESTAMP NOT NULL,  
  status VARCHAR(50) NOT NULL,
```

```

price DECIMAL(7,2) NOT NULL,
CONSTRAINT pk_booking_ddl PRIMARY KEY (booking_id),
CONSTRAINT fk_book_passenger FOREIGN KEY (passenger_id) REFERENCES
Passengers_DDL(passenger_id)
);

CREATE TABLE Baggage_check (
baggage_check_id INT NOT NULL,
check_result VARCHAR(50) NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
booking_id INT NOT NULL,
passenger_id INT NOT NULL,
CONSTRAINT pk_baggage_check PRIMARY KEY (baggage_check_id),
CONSTRAINT fk_bagchk_booking FOREIGN KEY (booking_id) REFERENCES
Booking_DDL(booking_id),
CONSTRAINT fk_bagchk_passenger FOREIGN KEY (passenger_id) REFERENCES
Passengers_DDL(passenger_id)
);

CREATE TABLE Baggage (
baggage_id INT NOT NULL,
weight_in_kg DECIMAL(4,2) NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
booking_id INT NOT NULL,
CONSTRAINT pk_baggage PRIMARY KEY (baggage_id),
CONSTRAINT fk_bag_booking FOREIGN KEY (booking_id) REFERENCES Booking_DDL(booking_id)
);

CREATE TABLE Boarding_pass (
boarding_pass_id INT NOT NULL,
booking_id INT NOT NULL,
seat VARCHAR(50) NOT NULL,
boarding_time TIMESTAMP NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
CONSTRAINT pk_boarding_pass PRIMARY KEY (boarding_pass_id),
CONSTRAINT fk_bp_booking_ddl FOREIGN KEY (booking_id) REFERENCES
Booking_DDL(booking_id)
);

CREATE TABLE Flights_DDL (
flight_id INT NOT NULL,
sch_departure_time TIMESTAMP NOT NULL,
sch_arrival_time TIMESTAMP NOT NULL,
departing_airport_id INT NOT NULL,
arriving_airport_id INT NOT NULL,
departing_gate VARCHAR(50) NOT NULL,
arriving_gate VARCHAR(50) NOT NULL,
airline_id INT NOT NULL,
act_departure_time TIMESTAMP NOT NULL,
act_arrival_time TIMESTAMP NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
CONSTRAINT pk_flights_ddl PRIMARY KEY (flight_id),
CONSTRAINT fk_fl_dep_airport FOREIGN KEY (departing_airport_id) REFERENCES
Airport(airport_id),
CONSTRAINT fk_fl_arr_airport FOREIGN KEY (arriving_airport_id) REFERENCES
Airport(airport_id),
CONSTRAINT fk_fl_airline FOREIGN KEY (airline_id) REFERENCES Airline_info(airline_id)
);

```

```
CREATE TABLE Booking_flight (  
  booking_flight_id INT NOT NULL,  
  booking_id INT NOT NULL,  
  flight_id INT NOT NULL,  
  created_at TIMESTAMP NOT NULL,  
  updated_at TIMESTAMP NOT NULL,  
  CONSTRAINT pk_booking_flight PRIMARY KEY (booking_flight_id),  
  CONSTRAINT fk_bf_booking FOREIGN KEY (booking_id) REFERENCES Booking_DDL(booking_id),  
  CONSTRAINT fk_bf_flight FOREIGN KEY (flight_id) REFERENCES Flights_DDL(flight_id)  
);
```

```
ALTER TABLE Airline_info RENAME TO Airline;
```

```
ALTER TABLE Booking_DDL RENAME COLUMN price TO ticket_price;
```

```
ALTER TABLE Flights_DDL ALTER COLUMN departing_gate TYPE TEXT;
```

```
ALTER TABLE Airline DROP COLUMN info;
```